DD-B-599a
December 1, 1961
SUPERSEDING
Fed. Spec. DD-B-599
January 29, 1952
(See 6.4)

FEDERAL SPECIFICATION

BOTTLE, STOPPER, VACCINE, ROUND AND STOPPER, BOTTLE, RUBBER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers round, vaccine, stopper bottles, and a rubber bottle stopper suitable for use therewith

1.2 Classification.

1.2.1 *Sizes* Rubber bottle stopper shall be of one size Vaccine bottles shall be of the following sizes, as specified (see 6 1):

Size I—5-milliliter (ml) capacity

Size II—10-ml capacity.

Size III—25-ml capacity

Size IV—50-ml capacity

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-621—Boxes, Wood, Nailed and Lock Corner.

PPP-B-636—Box, Fiberboard.

PPP-B-676—Boxes Set-Up, Paperboard

Federal Standards:

Fed Std No 102—Preservation, Packaging and Packing Levels

Fed Std No 123—Marking for Domestic Shipment (Civilian Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U S Government Printing Office, Washington 25, D C

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston New York Atlanta, Chicago Kansas City Mo , Dallas Denver, San Francisco, Los Angeles, Seattle, and Washington, D C

(Federal Government activities may obtain copies of Federal Specifications Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-L-10547—Liners, Case, Waterproof.

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Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing Materials Standard

ASTM Designation C 162–56—Definitions of Terms Relating to Glass and Glass Products.

(Copies of ASTM Standards may be obtained upon application to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

U. S. Pharmacopoeial Convention, Inc.:
Pharmacopoeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Glass.* Vaccine bottles shall be made of transparent, colorless, borosilicate glass.

3.1.1.1 *Chemical durability.* Glass shall be of such composition that when subjected to the test specified in 4.4.1, not more than 1.0 ml. of fiftieth-normal sulfuric acid shall be consumed.

3.1.1.2 *Resistance to thermal shock.* Bottles shall not break, crack, or chip when tested as specified in 4.4.2.

3.1.2 *Rubber.* Rubber stopper shall be made of a compound containing at least 75 percent by volume of new live rubber, colored with iron oxide or other inorganic nonbleeding coloring material, and compounded with a filler equal to at least 10 percent by weight of high-grade zinc oxide. The finished product shall be resilient, nonblooming, and free of tackiness.

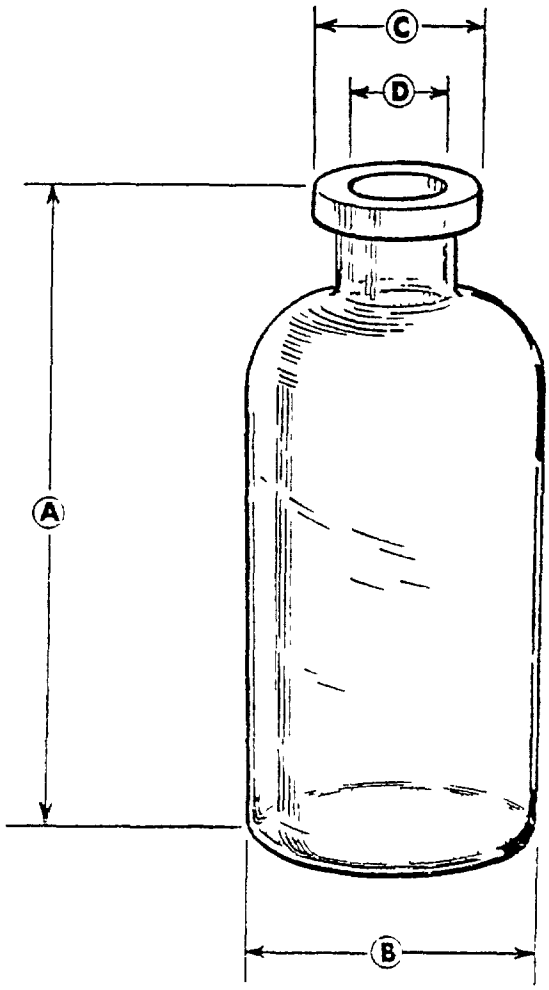
3.1.2.1 *Aging.* The stopper shall be free of tackiness, cracks, and other changes in condition when tested as specified in 4.4.3.

3.2 Style and design.

3.2.1 *Vaccine bottle.* The bottle shall be a conventional, cylindrical-shaped round-shouldered vaccine bottle, and shall conform to figure 1. The neck shall have a smooth cylindrical inside finish. It shall be so designed as to admit and hold the sleeve-type rubber stopper described in 3.2.2. Unless otherwise specified, the top of the neck of the bottle may have a bead with round edges. The bottle shall be furnished without the stopper. Dimensions and weight for vaccine bottles, sizes I through IV, shall be as indicated on figure 1.

3.2.2 *Rubber stopper.* Rubber stopper shall be commercial size No. 14, and shall conform to figure 2. Shoulder formed by the union of the stopper sleeve and core shall be rounded. Rubber stopper shall bear a circular imprint or other device to mark its center. The core shall be hollow and shall have straight sides (not tapered). The wall of the sleeve shall have straight sides, and its outer circumference shall form the outer circumference of the top of the stopper. Dimensions shall be as indicated on figure 2.

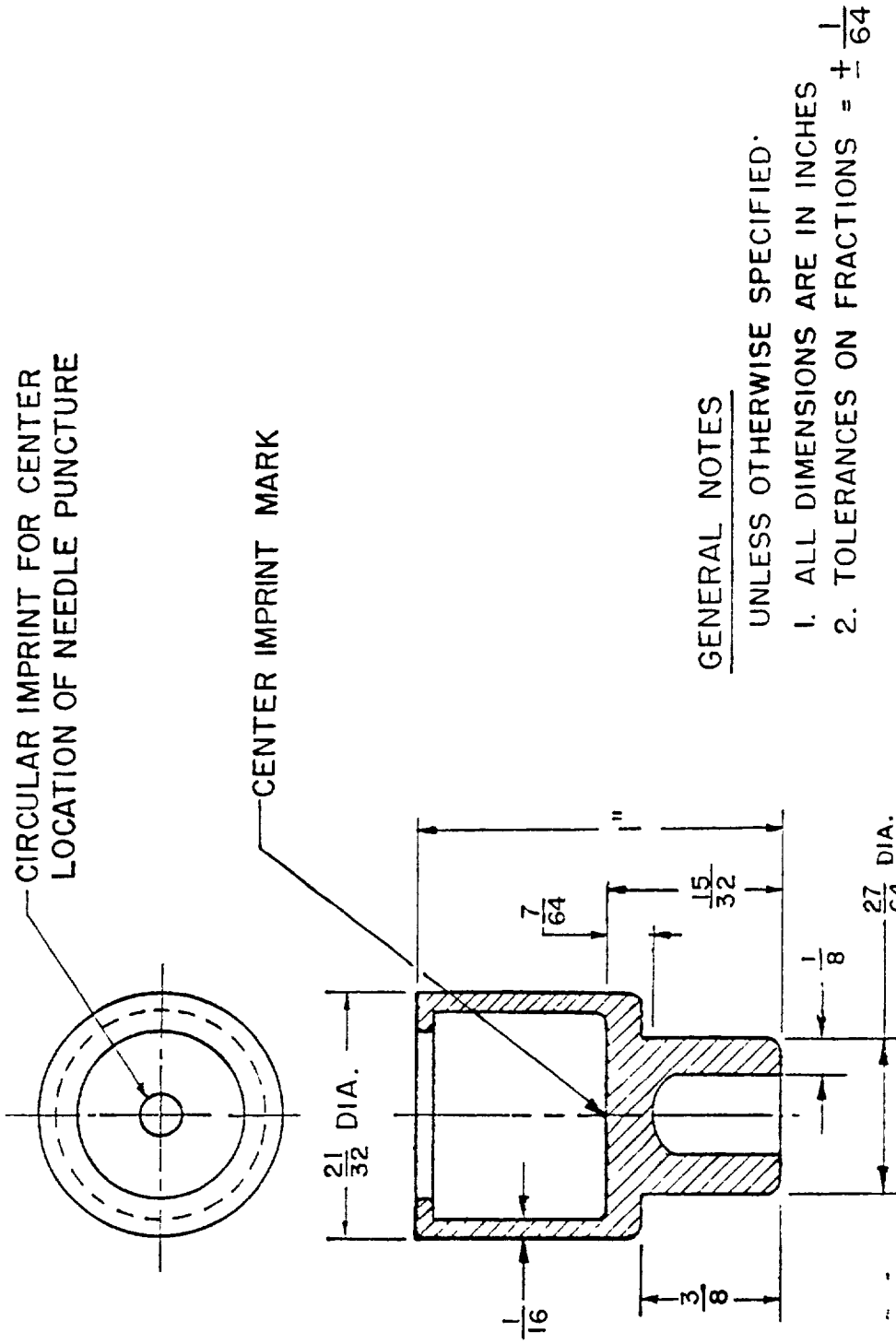
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SIZE	CAPACITY ML	Ⓐ HEIGHT OVER-ALL MM	Ⓑ DIAMETER OUTSIDE BODY MM	Ⓒ DIAMETER OUTSIDE BEAD MM	Ⓓ BOTTLE NECK INSIDE DIAMETER MM	WEIGHT GRAMS
I	5	46-52	21.9-23.8	18.7-20.3	9.00-10.05	12.6-22
II	10	61-63	27.2-28.8	18.7-20.3	9.00-10.05	22.5-27.5
III	25	76-78	32.2-33.8	18.7-20.3	9.00-10.05	36-44
IV	50	90-92	39.2-40.8	18.7-20.3	9.00-10.05	54-66

Figure 1

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STOPPER, BOTTLE, RUBBER
VACCINE, NO 14: WITH SLEEVE

FIGURE 2

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3.3 Identification marking. The name or registered trademark of the manufacturer shall be molded or otherwise permanently marked on the bottom of each bottle. In addition, each bottle shall have molded onto the outside surface, the capacity and capacity point.

3.4 Workmanship. Workmanship shall be first class throughout Bottles shall be symmetrical, and shall be free of irregularities and defects which affect their appearance or which may impair their serviceability Stoppers shall be symmetrically made and uniform in size and shape The surface shall be smooth and shall be free of dirt and foreign matter. The cut surface shall be of a uniform texture, free of pits and ingrained particles

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.1.1 *Inspection.* Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality* When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 **For examination.** Sampling for examination shall be conducted in accordance with Military Standard MIL–STD–105

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.5
Total (major and minor combined) defects .	II	4 0
For dimensional examination	L2	1 0

4.2.2 *For tests* Sampling for tests shall be conducted in accordance with Military Standard MIL–STD–105

Component (unit of product)	Characteristic	Requirement	Test procedure	Rqmt Appl: Indiv unit	Rqmt Appl: Lot avg	Inspection level	AQL (percent defective)
Bottle stopper	Chemical durability	3 1.1.1	4 4 1		λ	*	**
	Thermal shock ...	3 1 1.2	4 4.2	x		L3	4 0
	Aging	3.1.2.1	4 4.3	x		L3	**

* Sufficient sample of glass, selected in accordance with the Pharmacopoeia of the United States, representative of each lot or batch, shall be taken to perform one test as specified.

** Acceptance number shall be zero.

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4.2.3 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The bottles and stoppers shall be examined to determine compliance with all requirements contained in this specification

4.3.1 *Classification of defects* Examination shall be conducted in accordance with the following classification of defects:

*Categories and defects**

Major	
101	Bottle not free of cracks, breaks and chips
102	Bottle not free of blisters over 3/32 inch and stones over 1/32 inch
103	Square or round-edged bead at top of neck not furnished
104	Inside neck of bottle not smooth and not cylindrical
105	Shoulders of bottle not round
106	Bottle not of cylindrical shape
107	Walls of plug and sleeve o stopper not with straight sides
108	Circumference of outer sleeve wall of stopper not the same as that at top of stopper
109	New live rubber content in rubber compound for stopper not 75 percent by volume
110	Rubber compound for stopper not colored with iron oxide or other inorganic nonbleeding material
111	Rubber for stopper not compounded with a filler equal to at least 10 percent by weight of high-grade zinc oxide
112	Rubber stopper not furnished with a circular imprint or other means to mark center of stopper diaphragm
Minor	
201	Bottle not free of bubbles from 1/16 to 3/32 inch seeds stones, tears and sharp edges
202	Identification marking on bottle missing impermanent, or illegible
203	Bottle not stable on smooth or flat surface
204	Bottle and rubber stopper not symmetrical
205	Bottle and rubber stopper not free of dirt and foreign matter.
206.	Cut surface of rubber stopper not smooth, and not free of dirt and foreign matter.

* Examination shall not be restricted to the classified possible defects listed above Definitions of glass defects shall be in accordance with ASTM Designation C 162–56

4.3.2 *Dimensional examination* Bottles and stoppers shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect

4.4 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 *Chemical durability* The chemical durability of the bottles shall be tested in accordance with the procedures described in the latest revision of the Pharmacopoeia of the United States for type I glass

4.4.2 *Resistance to thermal shock* The specimens shall be tested by immersing them in water at 100° centigrade (C) ± 1°C, allowing them to remain for 5 minutes, and then transferring them to a water bath at a temperature of 23°-25°C, immersing them for 30 seconds and then removing them from the bath The 5-minute immersion in the hot bath shall be controlled within 3 seconds and the time taken for transfer from the hot to the room temperature bath shall be 15 + 0 33 seconds The water shall not be emptied from the specimens during the transfer from the hot to the room temperature bath During the tests the specimens shall be protected from drafts in a sheltered laboratory

4.4.3 *Aging* Stoppers shall be subjected to the action of oxygen under 300 pounds ± 10 pounds pressure per square inch at a temperature of 70° ± 1°C for 72 hours The stoppers shall then be allowed to stand for 24 hours at room temperature, pressure, and atmosphere before examining

5. PREPARATION FOR DELIVERY

For civil agency procurement only, Federal Standard No 102 should be referred to for definitions and applications of the various levels of packaging and packing protection for supplies and equipment.

5.1 Packaging.

5.1.1 *Unit of issue.* One bottle or stopper, as specified, constitutes one unit of issue.

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5.1.2 *Packaging quantities* The number of units of issue indicated in table I shall be packaged in each intermediate and exterior container, as applicable, for the required level of protection specified in the procurement document

TABLE I—*Packaging quantities*

Item	Size	Intermediate package	Exterior container
Bottle	1	144	1,152
Bottle . . .	II	144	864
Bottle . .	III	144	576
Bottle .	IV	144	288
Stopper .	14	500	9,000

5.1.2.1 *Packing variation permitted* If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design acceptable to a common carrier, which will insure safe delivery to destination

5.1.3 *Level A*

5.1.3.1 *Intermediate package*

5.1.3.1.1 *Bottles* Intermediate package shall be a carton of appropriate size and design, constructed in accordance with Federal Specification PPP-B-636 Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents Each carton shall be adequately secured to prevent accidental opening

5.1.3.1.2 *Stoppers* Intermediate package shall be a carton of appropriate size and design constructed in accordance with Federal Specification PPP-B-636 type I class 1, or PPP-B-676 Each carton shall be adequately secured to prevent accidental opening

5.1.4 *Level C* Bottles or stoppers shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.* Exterior container shall be constructed in accordance with Federal Specification PPP-B-621, table IV.

5.2.1.2 *Case liner* Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL-L-10547 Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping* Strapping, when required, shall be in accordance with applicable box specification

Note Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution

5.2.2 *Level B.*

5.2.2.1 *Exterior container* Exterior container shall be constructed in accordance with Federal Specification PPP-B-621, table I, or PPP-B-636, type I, class 1, fabricated from not less than 275-pound test fiberboard

5.2.3 *Level C* The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate to point of delivery called for in the contract or order

5.3 Marking—nonmilitary.

5.3.1 Unless otherwise specified in the contract or order, interior packages and exterior containers shall be marked in accordance with Federal Standard No 123

5.4 Marking—military.

5.4.1 *Interior packages.* Marking shall conform to the requirements of Military Standard MIL-STD-129.

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5.4.2 *Exterior container.* Exterior container shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following.

(*a*) Number and date of this specification

(*b*) Size of bottle required (see 1 2 1)

(*c*) Selection of applicable levels of packaging and packing (see 5.1 and 5 2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.3 Transportation description. Transportation descriptions applicable to these items are

Bottles, NOIBN, other than cut, one gallon or less in capacity, in boxes

Carload minimum weight 30,000 pounds in accordance with Uniform Freight Classification.

Truckload minimum weight 30,000 pounds in accordance with National Motor Freight Classification.

Bottle Stoppers, NOIBN, in boxes

Carload minimum weight 30,000 pounds in accordance with Uniform Freight Classification

Truckload minimum weight 30,000 pounds in accordance with National Motor Freight Classification.

6.4 Supersession data. This specification includes the requirements of Military Specification MIL–S–2633, dated 26 March 1951

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CUSTODIANS:

Army—M

Navy—Md

Air Force—AFCSC

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